

A4 · Where Does This Datagram Go?

CS425 Computer Networks · instructor led · every command runs on Onyx (`ssh onyx`)

TURN THIS IN ON PAPER
before you leave the room
graded pass/fail

Names _____ Date _____

Every round: **predict** first, in ink, then **run** the command with the projector, then **check**. A wrong prediction costs nothing. A prediction written afterwards is not a prediction.

ROUND 1 · ONYX'S ADDRESS `ip -4 -br addr`

QUESTION	ANSWER
Predict: addresses in Onyx's subnet, and how many can be given to hosts	
Onyx's address and prefix length, from the output	
Last octet of Onyx's address in binary, with a line after the prefix bits	
Network address · broadcast address	
Usable host range · number of usable hosts	
Is 132.178.227.200 on Onyx's subnet? Is Onyx's address private (RFC 1918)?	

Scratch. Work the binary here: the last octet of the address, the last octet of the mask, and the last octet of **.200**.

ROUND 2 · LONGEST PREFIX MATCH `ip route · ip route get`

DESTINATION	PREDICT: MATCHING LINE	PREDICT: DIRECT OR GATEWAY	WHAT <code>ip route get</code> SAID
132.178.227.50			
132.178.227.200			
8.8.8.8			
127.0.0.1			

Q1. **132.178.227.50** matches both lines of the table. Why does it not go to the gateway? Where did the answer for **127.0.0.1** come from, when it is not in `ip route`?

ROUND 3 · HOW BIG CAN A DATAGRAM BE? `ping -c 1 -M do -s SIZE 132.178.227.1`

QUESTION	ANSWER
MTU of eno12399np0, from <code>ip link show</code>	
Predict: the biggest <code>-s</code> payload that fits in one datagram	
What happened at <code>-s 1472</code> ? At <code>-s 1473</code> ?	

Q2. Where did the other 28 bytes go? Did `message too long` come from the network or from Onyx? Without `-M do`, who would fragment the datagram, and who would reassemble it?

ROUND 4 · COUNTING ROUTERS WITH TTL `tracpath -n mirrors.xmission.com`

QUESTION	ANSWER
Predict: routers between Onyx and Salt Lake City (about 475 km)	
Hops <code>tracpath</code> reported · how many said <code>no reply</code>	

Q3. How did `tracpath` learn the address of hop 3? The trace reached the end despite the `no reply` hops. What does that tell you about those routers?

Exit. A traditional router forwards on the destination address only. Name one other header field a middlebox on that path could match on, and what it might do with a match.

Graded pass/fail on something in every box. Hand this in before you leave.